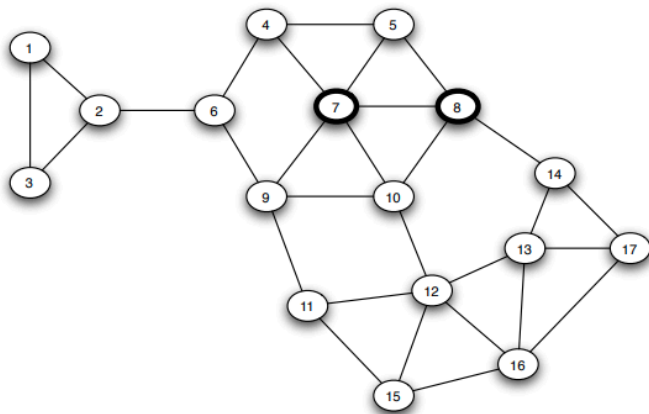
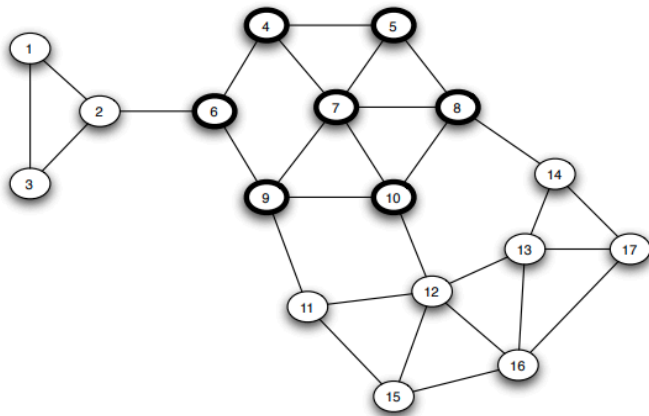


L10: Cascades in Networks with Communities-2



(a) Two nodes are the initial adopters



(b) The process ends after three steps

Image source: Textbook "[Networks, Crowds and Markets](#)"

(<https://www.cs.cornell.edu/home/kleinber/networks-book/networks-book-ch19.pdf>) book, by Easley and Kleinberg, Chapter 19

Suppose that we model the diffusion process with the Linear Threshold Model. To keep things simpler, let us consider the homogeneous version of this model in which all edge weights are the same.

If the threshold is θ then a node becomes active if and only if at least a fraction θ of its neighbors are active.

The visualization shows an example in which the cascade starts from two nodes (nodes 7 and 8). If the threshold θ is larger than $2/3$, the cascade will not expand beyond those initial nodes.

If $\theta = 2/5$, then the cascade in this example will expand to the seven nodes that are highlighted within 3 steps. In the first step, nodes 5 and 10 will become active. In the second step, nodes 4 and 9 will become active. And in the third step, node 6 will become active. The cascade will stop at that point.

Note that these 7 highlighted nodes form a cluster of density $p=2/3$.

Is there a relation between the density of the cluster p , and the maximum value of the activation threshold θ that is required for the establishment of a complete cascade in the network?

This is the main result we will establish on the next page.